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14.

	$x \rightleftharpoons y + z$			$A \rightleftharpoons 2B$	
	x	y	z	A	B
Initial no. of moles	a	–	–	a	–
Number of moles dissociated	x	–	–	x	–
No. of moles at equilibrium	a – x	x	x	a – x	2x
Total no. of moles at equilibriant	a – x + x + x + a + x			a – x +2x a + x	

$$\frac{K_{P_1}}{K_{P_1}} = \frac{P_y P_z}{P_x} \times \frac{P_A}{P_B^2}$$

$$\frac{K_{P_1}}{K_{P_2}} = \left[\frac{\left(\frac{x}{a+x}\right)P_1\left(\frac{x}{a+x}\right)P_1}{\frac{(a-x)}{(a+x)} \times P_1}\right] \times \left[\frac{\frac{(a-x)}{(a+x)}P_2}{\frac{4x^2P_2^2}{(a+x)^2}}\right]$$

$$\frac{K_{P_1}}{K_{P_1}} = \frac{P_1}{4P_2}$$

Given that $\frac{K_{P_1}}{K_{P_2}} = \frac{9}{1}$

$$\therefore \frac{9}{1} = \frac{P_1}{4P_2} \Rightarrow \frac{P_1}{P_2} = \frac{36}{1}$$

option (a) is correct

15. $K_C = \frac{[Fe^{3+}][OH^-]^3}{[Fe(OH)_3(s)]}$

$$K_C = [Fe^{3+}] \times \frac{1}{64}[OH^-]^3$$

To maintain K_C as constant, concentration of Fe^{3+} will increase by 64 times.

16. $K_p = 0.5$

$$Q = \frac{P_{PCl_3} \cdot P_{Cl_2}}{P_{PCl_5}}$$

$$Q = \frac{1 \times 1}{1}$$

$Q > K_p$ ∴ Reverse reaction is favoured ; i.e. more PCl5 will be produced. option (c)

17. V = 1L

$H_2 + I_2 \rightleftharpoons 2HI$

$[H_2]_{initial} = [I_2]_{initial} = a$

$[H_2]_{eq} = [I_2]_{eq} = (a - x)$

and $[HI]_{eq} = 2x$

$$K_C = \frac{[HI]^2}{[H_2][I_2]}$$

$$\therefore K_C = \frac{4x^2}{(a-x)^2}$$

Given that $K_C = \frac{K_f}{K_r} = 1$

∴ $4x^2 = (a - x)^2$

$4x^2 = a^2 + x^2 – 2ax$

$3x^2 + 2ax – a^2 = 0$

$x = –a$ & $x = \frac{a}{3}$

degree of dessociation = $\frac{a}{3} \times 100$

= 33.33 %

option (a)

18. $K_f = 2.5 \times 10^2$

$K_C = 50$ $K_r = ?$

$$K_C = \frac{K_f}{K_r}$$

$$50 = \frac{2.5 \times 10^2}{K_r}$$

$K_r = 5$ (option b)

19. correct statement : Physical processes occurs at the same rate at equilibrium

∴ option (c) is incorrect statement

20. $2SO_2 + O_2 \rightleftharpoons 2SO_3$

$$K_1 = \frac{[SO_3]^2}{[SO_2]^2[O_2]}$$

dissociation of 1 mole of SO_3